

Interview ID	Question number	Question	Short answer	Relevant codes	Key quote or paraphrase	Follow-up needed?
INT-01	Q1	Which window components usually fail first in residential window systems?	Hardware, opening mechanisms and handles usually fail or wear first. Glazing units may become a later problem area.	Physical obsolescence, Technical obsolescence, Maintenance, Component replacement	Early failures often occur in shorter-life components rather than in the full frame system.	No
INT-01	Q2	Which failures can usually be repaired without replacing the complete window?	Hardware, handles and glazing units can usually be repaired or replaced without replacing the whole window.	Repair, Component replacement, Component accessibility, Maintenance	Component-level repair can extend window life when access and compatible parts are available.	No
INT-01	Q3	What prevents repair when repair is technically possible?	Poor access to hinges, mechanisms or other serviceable parts can prevent repair, especially when the window reveal or finishing is poorly executed.	Component accessibility, Repair information, Missing repair infrastructure, Skilled labour, Implementation gap	The window may be repairable in theory, but not easily repairable in practice.	No
INT-01	Q4	How long are compatible spare parts normally available?	Spare-part availability becomes more difficult for very old window systems, especially around 25 years or older.	Spare parts, Supplier commitment, Implementation gap, Technical obsolescence	Long-term repairability depends on continued availability of compatible parts.	No
INT-01	Q5	Why do clients choose full replacement instead of repair or maintenance?	Clients often choose full replacement because they want a newer, thicker, better or more modern window system.	Premature replacement, User acceptance, Aesthetic obsolescence, Functional obsolescence, Economic obsolescence	Replacement decisions are not only technical, but also shaped by expectations and perceived improvement.	No
INT-01	Q6	Can modularity or replaceable parts create extra cost, material use, technical complexity or maintenance problems?	The supplier connected cost mainly to market conditions and material prices, not to replaceable parts themselves.	Economic obsolescence, Higher initial material use, Lifecycle trade-off, User acceptance	Replaceable parts were not identified by the supplier as a direct cause of more difficult maintenance.	No
INT-01	Q7	What prevents removed windows or window components from being reused?	Project-specific dimensions make direct reuse difficult. Not all removed components can realistically be reused, and glass also ages.	Direct reuse, Disassembly, Service infrastructure, Certification, Technical obsolescence, Recycling	Theoretical reuse potential does not automatically become practical reuse.	No
INT-01	Q8	Which design decision most strongly influences future repairability?	Good access to components is the most important condition, especially access to hinges, hardware and serviceable parts.	Design intervention, Component accessibility, Repair information, Maintenance planning	If there is good access, many things can be repaired.	No
INT-01	Q9	Have you seen a supposedly sustainable window decision create an unexpected problem later?	The supplier did not provide a direct example of a sustainable or energy-efficient window decision creating an unexpected problem later.	No direct example, General unintended consequence not identified	This is absence of evidence from this interview, not proof that such problems do not occur.	No
INT-01	Q10	What information would help architects make better decisions about window repair, replacement, service life and environmental impact?	Architects should ensure good access to the window and its components. The supplier did not provide specific environmental-impact or verified service-life information.	Component accessibility, Repair information, Lifecycle verification, Design intervention, Implementation gap	Practical repair knowledge does not automatically include lifecycle environmental information.	No
INT-02	Q1	When specifying windows in residential projects, which components do architects expect to fail first?	Opening and closing mechanisms are expected to wear first, especially dual-mode opening mechanisms.	Physical obsolescence, Technical obsolescence, Functional obsolescence, Maintenance, Component replacement	This aligns with the supplier interview and confirms the importance of shorter-life functional components.	No

INT-02	Q2	Are separately replaceable components considered in practice?	Separate component replacement is considered very important, especially for glazing units, seals, handles, hinges and fittings.	Component replacement, Repair, Component accessibility, Design intervention, Maintenance planning	Repairability is both a supplier issue and an architectural specification issue.	No
INT-02	Q3	What information is missing when architects assess repairability or future replacement?	The issue is not only missing information, but the gap between known repairable solutions and common construction practice. Installation subframes or pre-frames may improve future replacement, but are not widely used.	Implementation gap, Component accessibility, Design intervention, Repair information, Technical obsolescence, Lock-in	Repairable solutions may exist, but are not commonly adopted in practice.	No
INT-02	Q4	Do suppliers provide enough information about spare parts, maintenance, service life and warranties?	Suppliers can usually provide information if it is specifically requested, but it may not automatically enter the design process.	Repair information, Spare parts, Supplier commitment, Lifecycle verification, Implementation gap	Information flow between supplier, architect and client matters.	No
INT-02	Q5	Why do clients or investors prefer full replacement or new products?	Repair and replacement are difficult when windows are embedded into the wall assembly without accessible installation strategies.	Component accessibility, Disassembly, Implementation gap, Lock-in, Premature replacement, Design intervention	Repairability depends on the building interface, not only on the product.	No
INT-02	Q6	Can high-performance or modular window systems create extra cost, material complexity or future limitations?	More advanced systems may involve higher cost, more material and more complex mechanisms, but may also offer longer life and better energy performance.	Higher initial material use, Increased material complexity, Lifecycle trade-off, Economic obsolescence, Upgrade, Lifecycle verification	The benefit depends on whether added cost, material and complexity are justified over the lifecycle.	No
INT-02	Q7	What prevents removed windows or components from being reused in another project?	Lack of standard window dimensions is the main barrier. Windows are usually custom-made for specific openings.	Direct reuse, Standardisation gap, Service infrastructure, Disassembly, Certification, Implementation gap	Made-to-measure dimensions limit direct reuse.	No
INT-02	Q8	Which design or specification decision most strongly affects future repairability?	Dimensional standardisation can improve future repairability, replacement, maintenance and possible reuse.	Design intervention, Component accessibility, Standardisation, Maintenance planning, Direct reuse, Repair information	Standardised or repeated dimensions may reduce customisation barriers.	No
INT-02	Q9	Has a sustainable-looking window decision created an unexpected problem later?	The architect did not provide a direct example and expressed confidence that good design and correct specification should prevent serious problems.	No direct example, Design intervention, Lifecycle verification, General unintended consequence not identified	This is not proof that unintended consequences cannot occur.	No
INT-02	Q10	What information would help architects make better decisions?	Clearer client briefs, better client understanding, legal standardisation and stronger requirements would support better decisions.	Client brief, Regulation, Standardisation, Policy intervention, Lifecycle verification, User guidance, Implementation gap	The issue extends beyond product data to the wider decision-making system.	No